Chapter 39

39 Microscopic Energy Transfer Considerations 2.4.6, 2.4.7

So far, our statements of the second law of thermodynamics have been very practical. They tell us what will and what won't happen. Like we won't be able to make an engine that is more efficient than a Carnot engine. But surely the reason for what happens and what doesn't has to do with the molecules and forces that act on them. We found that we could describe things like pressure and temperature in terms of what happened to the molecules. It is time to do tat for the second law of thermodynamics.

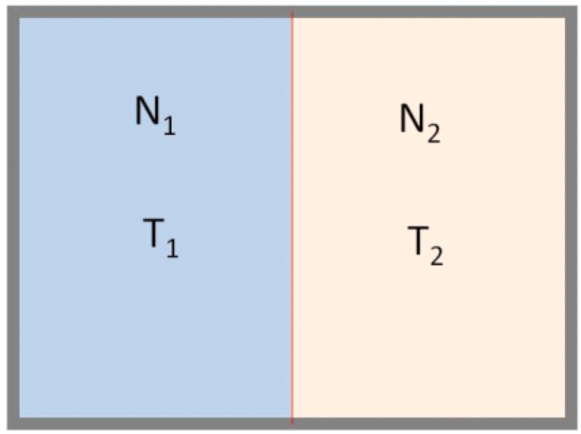
Fundamental Concepts

1. Microscopic idea of Energy Transfer
2. Microscopic idea of disorder
3. Entropy

39.1 Microscopic View of Energy Transfer

Suppose we have a chamber divided into two volumes. The volumes are separated by a membrane. If they have different temperatures, we expect that energy will transfer, but how?

Think of a basket ball court in a gym. In our gyms, we have large nets that divide courts to keep balls from passing from one court to another. Now think of millions of basket balls in each side of the gym net. The basket balls all are moving. Some of the basket balls will strike the net. If a ball on the other side strikes the net at a particular place and time , and a ball from the other side of the net strikes at the same place and time, the collision between the balls will transfer energy. This would be a conservation of momentum and energy problem The slower ball would gain energy, the faster ball would loose energy.



This is analogous to our gasses in chambers. The molecules, like the basketballs, transfer energy by collision.

The internal energy for each side of the chamber depends on both N, then number of molecules, and T, the temperature.

$$\begin{aligned} E_{1i} &= \frac{3}{2}N_1 k_B T_{1i} \\ E_{2i} &= \frac{3}{2}N_2 k_B T_{2i} \end{aligned}$$

The total energy is

$$E_{tot} = E_{1i} + E_{2i}$$

if our container does not change volume (no work done) and is insolated ($Q = 0$) then E_{tot} will not change. But eventually the two sides come to the same temperature

$$T_f = T_{1f} = T_{2f}$$

This happens when the average energy of the molecules (or basketballs) are the same. On average, neither side gains energy due to the collisions on the membrane (net). We can rewrite our energy equation for the final condition as

$$\bar{E}_{tot} = \bar{E}_{1f} = \bar{E}_{2f}$$

because the average energies will all be the same. Using

$$\begin{aligned} E_{1f} &= N_1 \frac{3}{2} k_B T_{1f} = N_1 \bar{E}_1 \\ E_{2f} &= N_2 \frac{3}{2} k_B T_{2f} = N_2 \bar{E}_2 \end{aligned}$$

We can rewrite this as an energy per molecule

$$\frac{E_{tot}}{N_1+N_2} = \frac{E_{1f}}{N_1} = \frac{E_{2f}}{N_2}$$

Which gives a set of expressions for the final energy of each side

$$\begin{aligned} E_{1f} &= \frac{N_1 E_{tot}}{N_1+N_2} \\ E_{2f} &= \frac{N_2 E_{tot}}{N_1+N_2} \end{aligned}$$

and we can verify that

$$\begin{aligned} E_{tot} &= E_{1f} + E_{2f} \\ &= \frac{N_1 E_{tot}}{N_1+N_2} + \frac{N_2 E_{tot}}{N_1+N_2} \\ &= E_{tot} \end{aligned}$$

Let's take what we have learned and mathematically express this as macroscopic formula and show it at a microscopic level as a problem example.

Let's show that if $Q_{tot} = 0$, and no work is done, then

$$Q_1 = -Q_2$$

This is a calorimetry problem! And we know how to do calorimetry problems.

Start with the first law,

$$\begin{aligned} Q_1 &= \Delta E_1 = E_{1f} - E_{1i} \\ Q_2 &= \Delta E_2 = E_{2f} - E_{2i} \end{aligned}$$

since no work is done. Let's take Q_1 first

$$Q_1 = E_{1f} - E_{1i}$$

since

$$E_{1f} = \frac{N_1 E_{tot}}{N_1+N_2}$$

and

$$E_{tot} = E_{1i} + E_{2i}$$

then

$$Q_1 = \frac{N_1\left(E_{1i}+E_{2i}\right)}{N_1+N_2} - E_{1i}$$

Then, applying some algebra,

$$\begin{aligned} Q_1 &= \frac{N_1\left(E_{1i}+E_{2i}\right)}{N_1+N_2} - E_{1i}\frac{N_1+N_2}{N_1+N_2} \\ &= \frac{N_1E_{1i} + N_1E_{2i} - E_{1i}N_1 - E_{1i}N_2}{N_1+N_2} \\ &= \frac{N_1E_{2i} - E_{1i}N_2}{N_1+N_2} \end{aligned}$$

At this point, let's play a mathematical trick Note that

$$0 = \frac{N_2 E_{2i}}{N_1 + N_2} - \frac{N_2 E_{2i}}{N_1 + N_2}$$

This seems like a funny way to write 0, but it works. Now let's take

$$Q_1 = \frac{N_1 E_{2i}}{N_1 + N_2} - \frac{E_{1i} N_2}{N_1 + N_2} + 0$$

Adding zero does not change the value of Q_1 so

$$\begin{aligned}
Q_1 &= \frac{N_1 E_{2i}}{N_1 + N_2} - \frac{E_{1i} N_2}{N_1 + N_2} + \frac{N_2 E_{2i}}{N_1 + N_2} - \frac{N_2 E_{2i}}{N_1 + N_2} \\
&= \frac{N_1 E_{2i}}{N_1 + N_2} + \frac{N_2 E_{2i}}{N_1 + N_2} - \frac{E_{1i} N_2}{N_1 + N_2} - \frac{N_2 E_{2i}}{N_1 + N_2} \\
&= \frac{N_1 E_{2i} + N_2 E_{2i}}{N_1 + N_2} - \frac{E_{1i} N_2}{N_1 + N_2} - \frac{N_2 E_{2i}}{N_1 + N_2}
\end{aligned}$$

The first term is just E_{2i}, and using $E_{tot} = E_{1i} + E_{2i}$ again we see

$$\begin{aligned}
Q_1 &= E_{2i} - \left(\frac{(E_{1i} N_2 + N_2 E_{2i})}{N_1 + N_2}\right) \\
&= E_{2i} - \left(\frac{N_2 (E_{1i} + E_{2i})}{N_1 + N_2}\right) \\
&= E_{2i} - \left(\frac{N_2 (E_{tot})}{N_1 + N_2}\right) \\
&= E_{2i} - E_{2f} \\
&= -Q_2
\end{aligned}$$

which is what we expect. We have shown our calorimetry equation using the ideas of microscopic energy transport for ideal gasses. This was interesting, but we still don't have a microscopic statement of the second law of thermodynamics. Let's consider motion of the molecules again.

39.2 The arrow of time.

Why does time go only one way? The answer to this question may surprise you. *We don't know.* There is nothing in our laws of motion that tells us that time should only go one way. Take motion under constant acceleration.

$$y_f = y_i + v_o \Delta t + \frac{1}{2} a \Delta t^2$$

and let's start with $t_i = 0$ so

$$y_f = y_i + v_o t_f + \frac{1}{2} a t_f^2$$

this equation describes the motion of an object under constant acceleration, like that of gravity near the Earth's surface. But nothing in the equation tells us that time must take on ever grater more positive values.

The truth is that there is only one physical law that tells us that time goes only one way, and that law is an empirical law that says essentially "time only goes one way." We experience time going one way, so we developed a law that says so. This is not a terribly convincing argument. But this is the state of physics.

Some processes (really all processes) can't actually run forward or backward. Idealized processes can be run in reverse, but not real process. We call processes that can only work one direction *irreversible processes*. If we shot a movie of an irreversible process and ran the movie backwards, it would be funny, because we would know that the reverse process can't happen.

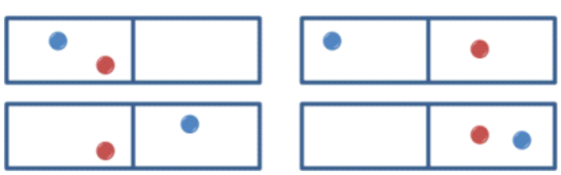
Let's take an example. Suppose our process is dropping a marker on the floor. Because the marker bounces are not perfectly elastic, the potential energy of the marker is converted into heat energy. If we added the lost energy back to the marker on the floor, it won't bounce up to a height for us to catch. This process of dropping the marker is irreversible.

To make this idea of irreversible processes and the arrow of time sound a little better, let's study thermodynamic situations and see that at least the arrow of time make sense. To do this, let's take a statistical approach (because we have learned that in thermodynamics we tend to use averages or rms values).

39.3 Disorder

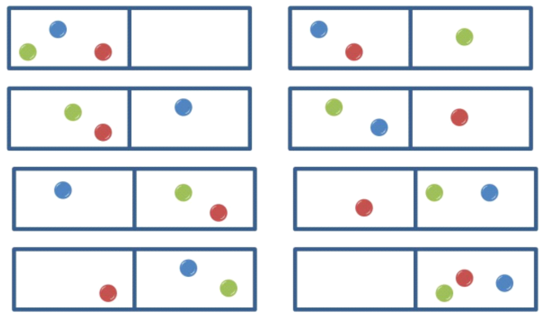
Suppose we have a system that has particles and places for those particles to be placed. We can start with a small space (see figure below). If I have one particle moving randomly, and divide my space into two compartments, then I have a probability of $\frac{1}{2}$ that the particle will be in the left compartment.

Now if I have two particles, the probability that two particles will be in the left compartment is $\left(\frac{1}{2}\right)^2$ or $\frac{1}{4}$.



We can think of this by looking at the ways we can put particles in the container. There are four ways to arrange the two particles in the two compartments. One of the four ways has both particles in the left compartment.

I can keep adding particles. For three particles the probability of being in the left side is $\left(\frac{1}{2}\right)^3$ or $\frac{1}{8}$.



For 50 particles it is

$$\left(\frac{1}{2}\right)^{50} = \frac{1}{1125\,899\,906\,842\,624}$$

We can extend this thinking to divisions of particles. Suppose I have 100 particles. We know that they will have a distribution of speeds (Maxwell-Boltzman distribution). What is the probability that the 50 fastest particles will be in the left side and the 50 slowest will be on the right side? This is the product of the last result with itself

$$\begin{aligned}\left(\frac{1}{2}\right)^{50}\left(\frac{1}{2}\right)^{50} &= \\ &= \frac{1}{1267\,650\,600\,228\,229\,401\,496\,703\,205\,376} \\ &= 7.888\,6 \times 10^{-31}\end{aligned}$$

This is not very probable. And we are only working with 100 particles and two sides of a box!

The point to this exercise is to realize there are very may *states* or ways to place particles in a divided box. Although each configuration is equally probable, suppose we want a particular outcome, say, the fastest 50 molecules on one side and the slowest 50 molecules on the other. There are many many more ways to *not* have this configuration that there are to have this configuration.

Each configuration of molecules and locations is called a *microstate*. There are a huge number of microstates in our example.

The collection of microstates is called a *macrostate*. There is more than one configuration of particles in our fast-slow division that places all the fast on one side and all the slow on another side. But the collection of all microstates that make up this division would be a macrostate. For example, we could have an even distribution of particles (within their fast/slow groups) on each side, or we could have all the fastest particles in each group on the top, and the slowest on the bottom within each major fast or slow side of the room. These two microstates sill preserve the division between the 50 fastest on the left and the slowest 50 on the right. They just rearrange the molecules within their groups.

39.4 Extension to large spaces and many particles

Now let's take a room full of molecules. We count the number of microstates by considering the number of places we can put a molecule. Each molecule has a volume $\not{V}_m$. So there are about

$$w = \frac{\not{V}}{\not{V}_m} \tag{39.1}$$

places to put the molecule that are distinct.[1] The w is not work, but we are reusing a symbol (again). It is the number of places that we can put a molecule.

Now suppose we have N molecules. If we ignore the probability that we might have two molecules in one space (which is different than the case above!) we have

$$(w)^N = \left(\frac{\not{V}}{\not{V}_m}\right)^N \tag{39.2}$$

ways to place these molecules among our w states. I'll call this

$$W = \left(\frac{\not{V}_i}{\not{V}_m}\right)^N \tag{39.3}$$

where $\not{V}_i$ is a particular volume (we will let it change to $\not{V}_f$ in just a minute).

39.5 A Measure of Disorder

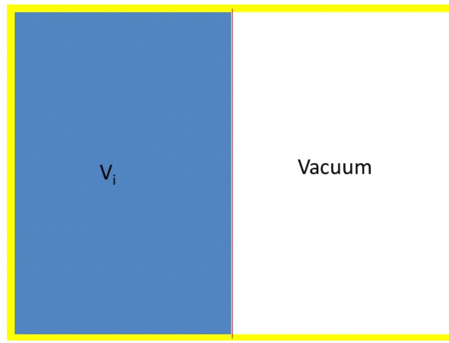
It would be convenient to find a way to measure the disorder. That is, we want to know how messed up the pattern of molecules in the room is. Then we could see how a thermodynamic process affects the amount of disorder. The traditional way to do this is to define a quantity

$$S \equiv k_B \ln(W) \tag{39.4}$$

This has the property that S increases as the number of states, W, increases. The logarithm is convenient because W is often very big for real systems and the $\ln(W)$ is a much smaller number to deal with.

Let's take an example, our old friend, the adiabatic free expansion.

[1] It's really more than this if you tight pack the molecules, like in a crystal structure. But this approximation is close enough for our study of the second law of thermodynamics.



For this case we start with all the atoms on one side of a membrane. No energy transfers out by heat. No work is done, we just puncture the membrane in the center and let the gas flow from one side to the other. We will mix up the molecules changing the amount of order. How does the disorder change in this process? Consider the number of states.

$$W_i = \left(\frac{\cancel{V}_i}{\cancel{V}_m}\right)^N$$

$$W_f = \left(\frac{\cancel{V}_f}{\cancel{V}_m}\right)^N$$

We can write our disorder measure as

$$S_i \equiv k_B \ln\left(W_i\right)$$

and

$$S_f \equiv k_B \ln\left(W_f\right)$$

Then the change is order is

$$\begin{aligned}
\Delta S &= S_f - S_i \\
&= k_B \ln\left(W_f\right) - k_B \ln\left(W_i\right) \\
&= k_B \ln\left(\frac{W_f}{W_i}\right)
\end{aligned}$$

If we look at the ratio of the number of states

$$\begin{aligned}
\frac{W_f}{W_i} &= \frac{\left(\frac{\cancel{V}_f}{\cancel{V}_m}\right)^N}{\left(\frac{\cancel{V}_i}{\cancel{V}_m}\right)^N} \\
&= \left(\frac{\cancel{V}_f}{\cancel{V}_i}\right)^N
\end{aligned}$$

so

$$\begin{aligned}\Delta S &= k_B \ln\left(\left(\frac{\cancel{V}_f}{\cancel{V}_i}\right)^N\right) \\ &= Nk_B \ln\left(\left(\frac{\cancel{V}_f}{\cancel{V}_i}\right)\right)\end{aligned} \tag{39.5}$$

We should ask if the disorder increased or decreased. Since $\cancel{V}_f$ is larger for our case than $\cancel{V}_i$, then ΔS will be a positive number. This means the disorder has increased in this case.

It is a good question to ask, will ΔS ever be negative? In other words, will the gas ever line up on only one side every again? If the answer is no, then this is an irreversible process. We already know that that this specific process is irreversible. We would have to do work to pump the gas back into one side. The gas won't spontaneously reorder itself.

Of course, we could set up a pump and pump our molecules back to one side of the container. That would lower the disorder, making ΔS negative. But now that we see that disorder is a function of molecules and their energy, think of the molecules around the pump. The pump has waste heat Q_c. That waste heat will add energy to the air molecules. That energy will be kinetic energy and now we have stirred up the air around the pump. We have increased the disorder for the pump environment. If we add $\Delta S_{container} + \Delta S_{environment}$ we will find that the total change in disorder for all systems is always positive or zero.

$$\Delta S \geq 0$$

39.6 Definition of entropy

We have learned to use one state variable, ΔE_{int}. The trick is to realize that ΔE_{int} only depends on ΔT and use any easy path to calculate it's value. That is why we can use

$$\Delta E_{int} = nC_V \Delta T$$

no matter what process we have. State variables, like ΔE_{int} give us powerful ways to calculate what will happen in a system.

There is another state variable in thermodynamics, *change in entropy*, ΔS. We will again find that we can calculate ΔS by any path because it depends only on the starting and ending points. And from our previous discussion we recognize that this new state variable, it is just a change in our disorder parameter!

$$\Delta S = k_B \ln\left(\frac{W_f}{W_i}\right)$$

Note that this is a difference between two disorder parameters. We will call our disorder parameter the *entropy* of the system.

$$S \equiv k_B \ln(W) \tag{39.6}$$

Entropy was originally defined at the macroscopic level. Statistical Mechanics (the microscopic study of thermodynamics) gives us our modern definition we have followed. We can state this definition as

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Entropy is a measure of the amount of disorder in a system and its environment.

39.6.1 Entropy and the Second Law

In our example, because, $\not{V}_f > \not{V}_i$ we found that ΔS will be positive. This implies that the disorder has increased. We can see that this is true, because there are now molecules distributed among many more states (position, occupation combinations) than before.

We will find this to be true in general, for irreversible processes (all natural process that are recognized). The total entropy of an isolated system that undergoes a change cannot decrease.

What about non-isolated systems? We must then consider the entropy of the system *and* its surroundings. We again find that entropy of a system *and* its surroundings cannot decrease. But the entropy of a system can decrease if the entropy of the surroundings increases more so the total effect is an increase in entropy (Think of the pump in our last example).

These statements constitute the most commonly seen form of the second law of thermodynamics.

Second Law of Thermodynamics: The entropy of a system and its surroundings increases over time.

$$\Delta S \geq 0$$

39.6.2 Macroscopic view

We don't really want to count the microstates of a process and look for changes in their number to do calculations. It would get tedious very fast! And so far our definition of ΔS was done only for a free expansion. We expect that in some way disorder would increase if we transferred in energy by heat, or did some kind of work on the system. We need a macroscopic version of our definition of entropy.

Let's look at the definition and then show that it works. The change in entropy is given by

$$\Delta S = \frac{\Delta Q_r}{T} \tag{39.7}$$

where Q_r is the energy transferred by heat through a reversible path and T is the absolute temperature (K) of the system. The assumption of a single temperature is because we have assumed ΔQ_r is small. But why do we use a reversible path?

Suppose we have an irreversible process. It might be hard to compute Q. But remember that, like ΔE_{int}, ΔS does not depend on path! It depends only on the initial and final states. So we may calculate it using any convenient path, and a reversible path is more convenient.

Note that macroscopically we have not defined entropy but the change in entropy, ΔS. This is the meaningful quantity (like ΔE_{int}) for thermodynamic calculations.

Let's do an example: What is the change in entropy when a block of ice melts? We will use

$$\Delta S = \int \frac{dQ_r}{T}$$

Note that while the ice is melting, we will have ice water at a constant $T_{melt} = 273\,\mathrm{K}$. Then our integral is

$$\Delta S = \frac{1}{T_{melt}} \int dQ_r$$

and the integral is not too hard

$$\Delta S = \frac{Q_r}{T_{melt}}$$

and for melting we know what Q_r could be. We can freeze water as well as melt ice. So we could model this as a reversible process. The amount of energy transfer needed to melt the ice is just

$$Q = mL$$

where m is the mass of the ice and L is the latent heat of fusion for water. So

$$\Delta S = \frac{mL}{T_{melt}}$$

This wasn't hard at all!

Let's do another example to tie our macroscopic ΔS to our microscopic ΔS. Let's find the change in entropy for a gas brought through an isothermal expansion. Again we have

$$\Delta S = \int_{V_i}^{V_f} \frac{dQ_r}{T}$$

and since the process is isothermal, T is constant

$$\Delta S = \frac{1}{T} \int_{V_i}^{V_f} dQ_r$$

and the integral would seem easy. It is just Q, but it is a particular Q for an isothermal process. For an isothermal process we know

$$\Delta E_{int} = 0$$

and

$$\Delta E_{int} = Q + w$$

so

$$Q = -w$$

and further we know that for an isothermal process

$$\begin{aligned} w &= -\int P d\cancel{V} \\ &= -nRT \ln\left(\frac{\cancel{V}_f}{\cancel{V}_i}\right) \end{aligned}$$

so the Q_r for our process could be just

$$Q = nRT \ln\left(\frac{\cancel{V}_f}{\cancel{V}_i}\right)$$

Then

$$\begin{aligned} \Delta S &= \frac{1}{T} nRT \ln\left(\frac{\cancel{V}_f}{\cancel{V}_i}\right) \\ &= nR \ln\left(\frac{\cancel{V}_f}{\cancel{V}_i}\right) \end{aligned}$$

and remember that $nR = Nk_B$ so

$$\begin{aligned} \Delta S &= \\ &= Nk_B \ln\left(\frac{\cancel{V}_f}{\cancel{V}_i}\right) \end{aligned}$$

which is just what we found from our microscopic view of entropy change! Now we realize that when we let the gas flow from one side of our divide box to the other, we did this isothermally. And that is true, the gas did not change the average kinetic energy per molecule as it flowed from one side of the box to the other. Since our system was isolated, $Q = 0$ and $W = 0$ for the whole system, so $\Delta E_{int} = 0$ and the system did not change temperature. But surely something changed! And that something was ΔS!

39.6.3 Heat death of the Universe

We should ask the question, is the universe an isolated system? We don't know the answer to this for sure[2], but we believe this is true that the universe is

[2] After all, Spiderman and his colleagues seem to be able to come and go from different "universes" all the time.

isolated.[3] That means that the change in entropy of the universe will always be positive. The universe is "running down." The ultimate end of this increase in energy would mean that all mater in the universe will be spread out and cold. This is called the *heat death of the universe*.

Note that this is a little different than the eschatological view that we have in the Church of Jesus Christ of Latter Day Saints.

Throughout my career, many colleagues and acquaintances have said it is foolish to believe in a church and a resurrection because the second law of thermodynamics tells us that the universe will die in a cold blur. How do we respond?

I can only give my opinion. But I think it is worth discussing, so here is a disclaimer for what follows.

WARNING: This is not an official statement of doctrine This is an answer to a question that many scientists must face from colleagues who will question their faith using the principals of thermodynamics. Official statements by the General Authorities supersede any opinions expressed here.

39.6.4 The First Law and LDS Thought[4]

I suppose we should ask, do we believe in the first law of thermodynamics? If this law does not work for us, we have no reason to expect the second law to work. The first law is basically conservation of energy. Here are some quotes to consider.

> And there stood one among them that was like unto God, and he said unto those who were with him: We will go down, for there is space there, and we will take of these materials, and we will make an earth whereon these may dwell; (Abraham 3:24)
>
> You ask the learned doctors why the say the world was made out of nothing; and they will answer, 'Doesn't the Bible say He created the world?' And they infer, from the word create, that it must have been made out of nothing. Now, the word create came from the word baurau which does not mean create out of nothing; it means to organize; the same as a man would organize materials and build a ship. (Joseph Smith, King Follett Discourse)

The idea of the first law of thermodynamics seems in harmony with LDS thought.

So let's take on the Second Law.

[3] Dispite what the television shows seem to say.

[4] Comparing physical theory to LDS thought has it's dangers, I am aware. But as scientists, we can't escape thinking about this. Just as a caution, John A. Widsoe wrote a book called *Joseph Smith as Scientist* in which Widsoe tried very hard to show that LDS though is in harmony with Universal Ether Theory. But we now know Universal Ether Theory is not correct, so being in harmony with it is meaningless. So we should be cautious!

39.6.5 The Second Law and LDS Thought

Again here are some quotes

> Wherefore, the first judgment which came upon man must needs have remained to an endless duration. And if so, this flesh must have laid down to rot and to crumble to its mother earth, to rise no more. (2 Nephi 9:7)
>
> And our spirits must have become like unto [Satan], and we become devils, angels to a devil, to be shut out from the presence of our God, and to remain with the father of lies, in misery, like unto himself. (2 Nephi 9:9)

So our bodies fall apart, with the molecules spreading to more and more states. And even our spirit seems to go occupy some kind of low energy state. We seem to accept the concept of the second law as well! –This will be a surprise to our non-member friends.

But then how can we expect eternal life and exaltation? Does something counter the second law (a Third or Fourth Law of Thermodynamics)?

An Additional Law of Thermodynamics?

Again let's go to the scriptures. Jacob describes another force not recognized by current scientific theory that counteracts this tendency to decay

> Wherefore, it must needs be an infinite atonement—save it should be an infinite atonement this corruption could not put on incorruption. Wherefore, the first judgment which came upon man must needs have remained to an endless duration. And if so, this flesh must have laid down to rot and to crumble to its mother earth, to rise no more. (2 Nephi 9:7-8)
>
> O how great the goodness of our God, who prepareth a way for our escape from the grasp of this awful monster; yea, that monster, death and hell, which I call the death of the body, and also the death of the spirit. (2 Nephi 9:10)

Atonement means to bring together. Like putting back all the atoms into on side of our container after they have expanded, or like putting a body back together after it decays, or putting a spirit back together with God. This is an apt description of a process or law that would counter or balance the second law[5]. Since the atonement can only be accessed through Christ, it is not likely to be discovered in a laboratory, but is a very real, physical, process. (see for example Luke 24:39)

This is not doctrine, but is a way to think about the second law until we get direct revelation on the subject. We really should not be worried about a

[5] Nibley, The meaning of the temple, reprinted in the Collected Works of Hugh Nibley: volume 12, Deseret Book, Salt Lake City, UT, 1992

conflict between thermodynamics and our doctrine. I don't believe one exists as you can see. It is totally OK for us (our bodies and spirits) to have a lower entropy while the surrounding universe has a larger entropy. The second law is not violated.

39.7 Last Thoughts

We are done with our introductory study of thermodynamics. Many of you will have junior level thermodynamics classes, so you are likely to see more of this. But we can stop for now. For physics majors the next physics class is likely to be PH220, electricity and magnetism. I would prefer to call this class "introduction to field theory" because we really study electric and magnetic fields for the first time. We will take up the topic of wave motion at the end of PH220 to mathematically describe light as electromagnetic waves. And then we will describe everything using waves in PH279, "modern physics," as our first introduction to quantum mechanics.

I hope you have enjoyed learning about waves, optics, and thermodynamics. I hope you will see the world a little differently (and think about the process of seeing the world!). For physics majors, there is still so many cool things to learn. If you are leaving physics at this point, you can continue to learn about the world and the universe by reading and watching physics related material. If you are innovating new techniques or products, you might even want to hire the services of a physicist to help with the underlying principles for your new venture!